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United States Patent	12398595
Kind Code	B2
Date of Patent	August 26, 2025
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Climbing product having an extendable section lock assembly

Abstract

A climbing product whose sections are extendable having a base section and a fly section. The fly section is in sliding engagement with the base section. The climbing product having a lock assembly having a lock fly portion directly attached to the fly section. The lock fly portion directly engages with the base section to lock the base section with the fly section. The lock fly portion disengages from the lock base portion by simply moving the fly section relative to the base section to unlock the fly section from the base section. A method for using a climbing product whose sections are extendable. A method for producing a climbing product whose sections are extendable.

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Appl. No.: 17/387856

Filed: July 28, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220034164 A1	Feb. 03, 2022

Related U.S. Application Data

us-provisional-application US 63140599 20210122
us-provisional-application US 63058805 20200730

Publication Classification

Int. Cl.: E06C1/12 (20060101); E06C7/04 (20060101); E06C7/06 (20060101)

U.S. Cl.:

CPC E06C1/12 (20130101); E06C1/125 (20130101); E06C7/04 (20130101); E06C7/06 (20130101);

Field of Classification Search

CPC: E06C (1/125); E06C (7/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
524784	12/1893	Monroe	N/A	N/A
2680555	12/1953	Nilsson	N/A	N/A
2900041	12/1958	Leavitt	N/A	N/A
3464522	12/1968	Kramer	N/A	N/A
3768594	12/1972	Kramer	N/A	N/A
4226302	12/1979	Roche	N/A	N/A
4605100	12/1985	Inglese	N/A	N/A
4754843	12/1987	Anderson	N/A	N/A
5000289	12/1990	Sanchez, III	N/A	N/A
10975622	12/2020	Yoo	N/A	N/A
2015/0047927	12/2014	Nodarse	N/A	N/A
2016/0123079	12/2015	Ballard	182/207	E06C 7/46
2020/0040656	12/2019	Mora	N/A	E06C 1/12
2022/0034163	12/2021	Lentine	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202125240	12/2011	CN	N/A
106837155	12/2016	CN	N/A
107075906	12/2016	CN	N/A
108386126	12/2017	CN	N/A
110792388	12/2019	CN	N/A
2412073	12/1974	DE	N/A
19507851	12/1994	DE	N/A
1070313	12/2008	ES	N/A
408298	12/1933	GB	N/A
1115672	12/1967	GB	N/A
2011163012	12/2010	JP	N/A
2019237077	12/2018	WO	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This is a nonprovisional of U.S. provisional patent application Ser. No. 63/058,805 filed Jul. 30, 2020 and 63/140,599 filed Jan. 22, 2021, all of which are incorporated by reference herein.

FIELD OF THE INVENTION

(2) The present invention pertains to a climbing product having an extendable section lock assembly. (As used herein, references to the “present invention” or “invention” relate to exemplary embodiments and not necessarily to every embodiment encompassed by the appended claims.) More specifically, the present invention pertains to a climbing product having an extendable section lock assembly where the lock assembly includes a fly lock portion attached to a fly section of the climbing product which locks with a base section of the climbing product, and which unlocks from the base section simply by the fly section being lifted relative to the base section.

BACKGROUND OF THE INVENTION

(3) This section is intended to introduce the reader to various aspects of the art that may be related to various aspects of the present invention. The following discussion is intended to provide information to facilitate a better understanding of the present invention. Accordingly, it should be understood that statements in the following discussion are to be read in this light, and not as admissions of prior art.

(4) Current locking systems for extendable ladders and other devices with extendable sections require the user to perform a separate action to unlock the extendable sections prior to moving them. The performance of a separate action to unlock the extendable sections can be burdensome and introduce possible errors that could cause the extendable sections to move relative to each other in an undesired manner, resulting in damage or injury. It would be desirable to eliminate the need to perform a separate action to unlock the extendable sections prior to moving.

(5) In addition, current locking systems on extension ladders are large and intrusive into a user's working space. It would be desirable to reduce or eliminate intrusion of a locking system into the user's working space.

BRIEF SUMMARY OF THE INVENTION

(6) The present invention pertains to a climbing product whose sections are extendable. The climbing product comprises a base section having a first base rail and a second base rail in parallel and spaced relationship with the first base rail and base rungs attached to and between the first and second base rails. The climbing product comprises a fly section having a first fly rail and a second fly rail in parallel and spaced relationship with the first fly rail and fly rungs attached to and between the first and second fly rails. The fly section is in sliding engagement with the base section. The climbing product comprises a lock assembly having a lock fly portion directly attached to the fly section. The lock fly portion directly engages with the base section to lock the base section with the fly section. The lock fly portion disengages from the lock base portion by simply moving the fly section relative to the base section to unlock the fly section from the base section.

(7) The present invention pertains to a method for using a climbing product **10** whose sections are extendable. The method comprises the steps of sliding a fly section **20** of the climbing product **10** relative to a base section **12** of the climbing product **10** to which the fly section **20** is in sliding engagement. There is the step of locking the fly section **20** to the base section **12**. There is the step of lifting the fly section **20** relative to the base section **12** to unlock the fly section **20** from the base section **12**. There is the step of moving the fly section **20** down relative to the base section **12**.

(8) The present invention pertains to a method for producing a climbing product **10** whose sections

are extendable. The method comprises the steps of attaching a lock assembly **28** having a lock fly portion **30** directly to a rung of a fly section **20** of the climbing product **10**. There is the step of sliding the fly section **20** along a base section **12** of the climbing product **10** so the fly section **20** engages with the base section **12**.

(9) The present invention pertains to a climbing product whose sections are extendable. The climbing product comprises a base section having a first base rail and a second base rail in parallel and spaced relationship with the first base rail and base rungs attached to and between the first and second base rails. The climbing product comprises a fly section having a first fly rail and a second fly rail in parallel and spaced relationship with the first fly rail and fly rungs attached to and between the first and second fly rails. The fly section in sliding engagement with the base section. The climbing product comprises a lock assembly having a lock fly portion directly attached to the fly section. The lock fly portion directly engages with the base section to lock the base section with the fly section. The lock fly portion has a plunger. The lock fly portion disengages from the base portion by moving the plunger.

(10) The present invention pertains to a method for using a ladder whose sections are extendable. The method comprises the steps of moving the ladder to a desired position. The ladder comprises a base section having a first base rail and a second base rail in parallel and spaced relationship with the first base rail and base rungs attached to and between the first and second base rails. The ladder comprises a fly section having a first fly rail and a second fly rail in parallel and spaced relationship with the first fly rail and fly rungs attached to and between the first and second fly rails. The fly section in sliding engagement with the base section the ladder comprises a lock assembly having a lock fly portion directly attached to the fly section. The lock fly portion directly engages with the base section to lock the base section with the fly section. The lock fly portion has a plunger. The lock fly portion disengages from the base portion by moving the plunger. There is the step of moving the plunger so the lock fly portion disengages from base portion. There is the step of sliding the fly section along the base section to extend the ladder to a desired length. There is the step of locking the fly section to the base section with the lock portion at the desired length.

(11) The present invention pertains to a method for constructing a ladder whose sections are extendable. The method comprises the steps of attaching a lock fly portion of the lock assembly directly to a fly section of the ladder. The fly section having a first fly rail and a second fly rail in parallel and spaced relationship with the first fly rail and fly rungs attached to and between the first and second fly rails. The fly section in sliding engagement with the base section. The lock fly portion has a plunger. There is the step of attaching a lock base portion of the lock assembly directly attached to the base section. The lock fly portion directly engages with the lock base portion to lock the fly section to the base section. The lock fly portion disengages from the lock base portion by moving the plunger. The base section having a first base rail and a second base rail in parallel and spaced relationship with the first base rail and base rungs attached to and between the first and second base rails.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 shows two sections of an extendable ladder.

(2) FIG. 2 shows the two sections assembled together with the moving section located within the rails of the fixed section in a typical MT ladder fashion.

(3) FIG. 3 shows two embodiments of the lock assembly of the present invention.

(4) FIG. 4 shows a first embodiment of a flip lock with the flip lock in the position to which it returns when free.

(5) FIG. 5 shows the flip lock is deflected as the base rung and fly rung begin to pass each other.

(6) FIG. 6 shows the flip lock as it enters a socket.

(7) FIG. 7 shows the flip lock wedged in the socket, which prevents the fly rung from further descending.

(8) FIG. 8 shows raising the fly rung withdraws the flip lock from engagement with the socket.

(9) FIG. 9 shows the flip lock is returned to its spring-biased position after it has cleared the base rung.

(10) FIG. 10 shows the flip lock deflects from the base rung to allow the fly rung to freely descend.

(11) FIG. 11 shows a second embodiment of a flip lock with the lock and the retractor in the position to which they are spring biased when free of any base rung.

(12) FIG. 12 shows the lock is deflected by the base rung as the fly rung is lifted.

(13) FIG. 13 shows the lock has cleared the base rung in the retractor has been partially deflected by the base rung.

(14) FIG. 14 shows the lock has engaged the base rung to prevent further motion downward as the fly rung has been moved down and the retractor is clear of the base rung.

(15) FIG. 15 shows the retractor is deflected by the base rung as the fly rung is lifted and the lock is clear of the base rung.

(16) FIG. 16 shows both the lock and the retractor are free and clear of the base rung as the fly rung has been lifted past the base rung.

(17) FIG. 17 shows the retractor has fitted into the slot of the lock because the base rung has deflected the retractor as the fly rung is moved down, thus allowing the fly rung to the sender freely.

(18) FIG. 18 shows further details of the lock assembly for the first embodiment.

(19) FIG. 19 is a cross-sectional side view showing the flip lock stop and the frame stop and the flip lock when the flip lock is free and clear of any base rung.

(20) FIG. 20 is a cross-sectional side view of the flip lock when it has been deflected upwards by a base rung.

(21) FIG. 21 is a cross-sectional side view of the flip lock when it has been deflected downwards by a base rung.

(22) FIG. 22 shows the lock assembly of the second embodiment.

(23) FIG. 23 is a cross-sectional side view of the lock in the retractor and the frame stops and the lock stop and the retractor stop when the lock and the retractor are free and clear of any base rung.

(24) FIG. 24 is a cross-sectional side view of the lock and the retractor when they have been deflected upwards by a base rung.

(25) FIG. 25 is a cross-sectional side view of the lock and that retractor when they have been deflected downwards by a base rung.

(26) FIGS. 26 and 27 show a ladder known as an NGMF (next generation multi form) ladder.

(27) FIG. 28 is a detailed view showing the inner rung with the flip lock engaged with one of the outer rungs.

(28) FIG. 29 shows just the inner rung with flip lock and an outer rung.

(29) FIG. 30 shows another view of just the inner rung with flip lock.

(30) FIG. 31 shows a view of a portion of a typical outer rung with a recess for the flip lock to engage.

(31) FIG. 32 shows the components of this invention as seen if the inner rung were transparent.

(32) FIG. 33 shows a closeup of FIG. 32 with some additional components made invisible. The key parts of this invention are the flip lock, blocker and plunger.

(33) FIG. 34 shows the relationship between the flip lock and the blocker.

(34) FIG. 35 shows a portion of the flip lock.

(35) FIG. 36 shows the conditions under which the plunger is able to engage the plunger locking recess in the flip lock: the flip lock is horizontal and the blocker is rotated to its extreme CCW position relative to the flip lock.

- (36) FIGS. **37**, **38** and **39** show how the plunger (axis shown by +symbol) is prevented from engaging the plunger locking recess (shown as a dashed line) whenever the blocker is in its extreme CW position relative to the flip lock.
- (37) FIGS. **40** and **41** show the function of the lock lever and the plunger. When the right end of the lock lever is down as in FIG. **40**, the plunger is in its left or engaged position with the locking recess. When the right end of the lever is lifted as in FIG. **41**, the plunger is moved to its right or disengaged position from the locking recess.
- (38) FIG. **42** shows a cross-section of how the contact between the blocker against the outer rung positions the blocker CCW relative to the flip lock as seen in FIG. **36**.
- (39) FIG. **43** shows a cross-section of the situation whenever the flip lock is NOT engaged in the recess in an outer rung: the blocker prevents the plunger engaging the flip lock so the flip lock is free to move in the manner of the previously disclosed flip lock invention.
- (40) FIG. **44** shows a cross-section of the routing of the hoist rope between the base section and the fly section.
- (41) FIG. **45** shows the ladder in the fully extended straight extension ladder position from the non-climbing side.
- (42) FIG. **46** shows the ladder in the fully extended straight extension ladder position from the climbing side.
- (43) FIGS. **47A-47E** show multiple views of the lock lever and lock lever handle in place on the rung.
- (44) FIG. **48** is a perspective view of the ladder with the fly sections extended.
- (45) FIG. **49** is a perspective view of the flip lock with the blocker and the flip lock spring, which biases the blocker, in place in the flip lock and adjacent the blocker.

DETAILED DESCRIPTION OF THE INVENTION

(46) Referring now to the drawings wherein like reference numerals refer to similar or identical parts throughout the several views, and more specifically to FIGS. **1-3** thereof, there is shown a climbing product whose sections are extendable. The climbing product comprises a base section **12** having a first base rail **14** and a second base rail **16** in parallel and spaced relationship with the first base rail **14** and base rungs **18** attached to and between the first and second base rails **14**, **16**. The climbing product comprises a fly section **20** having a first fly rail **22** and a second fly rail **24** in parallel and spaced relationship with the first fly rail **22** and fly rungs **26** attached to and between the first and second fly rails **22**, **24**. The fly section **20** in sliding engagement with the base section **12**. The climbing product comprises a lock assembly **28** having a lock fly portion **30** directly attached to the fly section **20**. The lock fly portion **30** directly engages with the base section **12** to lock the base section **12** with the fly section **20**. The lock fly portion **30** disengages from the lock base portion **32** by simply moving the fly section **20** relative to the base section **12** to unlock the fly section **20** from the base section **12**. The climbing product **10** may include but not be limited to MT ladders, leaning ladders, and work platforms.

(47) The lock assembly **28** may include a lock base portion **32** directly attached to the base section **12**, as shown in FIGS. **3-10** and **18-21**. The lock fly portion **30** directly engages with the lock base portion **32** to lock the fly section **20** to the base section **12**. The lock fly portion **30** disengages from the lock base portion **32** by simply moving the fly section **20** relative to the base section **12** to unlock the fly section **20** from the base section **12**. The lock base portion **32** may include a socket **34** disposed in one of the base rungs **18**, and the lock fly portion **30** includes a flip lock **36** which fits into the socket **34** to lock the fly section **20** to the base section **12** so the fly section **20** cannot move downwards relative to the base section **12** when the fly section **20** is locked to the base section **12**. The flip lock **36** disposed in one of the fly rungs **26**. The flip lock **36** may have a stem **38** which fits into the socket **34** to lock the fly section **20** to the base section **12**, an anchor **40** from which the stem **38** extends, a flip axle **42** extending through the anchor **40** about which the anchor **40** with the stem **38** rotates, and a flip spring **44** which biases the stem **38** into a stable position

where the stem **38** extends essentially perpendicularly toward the first base rail **14** from the first fly rail **22**.

(48) The lock fly portion **30** may include a lock **46** which directly engages with one of the base rungs **18** to lock the fly section **20** to the base section **12** so the fly section **20** cannot move downwards relative to the base section **12** when the fly section **20** is locked to the base section **12**. See FIGS. **11-17** and **22-25**. The lock fly portion **30** disengages from the lock base portion **32** by simply moving the fly section **20** relative to the base section **12** to unlock the fly section **20** from the base section **12**. The lock fly portion **30** may include a retractor **48** disposed adjacent the lock **46** which engages the lock **46** and maintains the lock **46** in a position out of the way of the base rungs **18** as the fly section **20** is moved downwards relative to the base section **12**. The lock fly portion **30** may have a lock axle **50** extending through the lock **46** about which the lock **46** rotates and a retractor axle **54** extending through the retractor **48** about which the retractor **48** rotates. The lock **46** may have an anchor **40**, a first arm **58** extending from the anchor **40** and a second arm **60** which is in spaced relation with the first arm **58**, where the first arm **58** is longer than the second arm **60** and where the second arm **60** extends from the anchor **40**. The first arm **58** and the second arm **60** and the anchor **40** together formed essentially a C shape and define a slot **62** between the first arm **58** and the second arm **60**. The lock fly portion **30** may include a lock spring **64** which biases the lock **46** into a stable position where the first arm **58** and second arm **60** extend toward the first base rail **14**. The lock fly portion **30** includes a retractor spring **66** which biases the retractor **48** into a stable position where the retractor **48** extends toward the first base rail **14**.

(49) The present invention pertains to a method for using a climbing product **10** whose sections are extendable. The method comprises the steps of sliding a fly section **20** of the climbing product **10** relative to a base section **12** of the climbing product **10** to which the fly section **20** is in sliding engagement. There is the step of locking the fly section **20** to the base section **12**. There is the step of lifting the fly section **20** relative to the base section **12** to unlock the fly section **20** from the base section **12**. There is the step of moving the fly section **20** down relative to the base section **12**.

(50) The present invention pertains to a method for producing a climbing product **10** whose sections are extendable. The method comprises the steps of attaching a lock assembly **28** having a lock fly portion **30** directly to a rung of a fly section **20** of the climbing product **10**. There is the step of sliding the fly section **20** along a base section **12** of the climbing product **10** so the fly section **20** engages with the base section **12**.

(51) In the operation of the invention, the lock assembly **28** allows for extendable sections to be moved and locked without the need to reach a locking mechanism, such as when extending and retracting the top section of an MT ladder when set up in extension mode. Furthermore, the lock assembly **28** provides for reduced or eliminated intrusion into the user's working space by being positioned mostly in or under rungs of the climbing product **10**.

(52) FIG. **1** shows two sections of an extendable ladder. These sections could be half of an MT type ladder or a straight extendable ladder. The fixed section, such as a base section **12**, rests on the ground while the moving section, such as a fly section **20**, is adjustable for different heights.

(53) FIG. **2** shows the two sections assembled together with the moving section located within the rails of the fixed section in a typical MT ladder fashion. It is desired that as the moving section is adjusted relative to the fixed section, it can be locked in various positions so that the rungs on the moving section are aligned with the rungs on the fixed section as shown.

(54) The typical locks used on MT ladders are known as J-locks. The two J-locks must be manually unlocked one at a time, the moving section moved to the desired position, then the J-locks manually locked one at a time. J-locks are typically located at the upper end of the fixed section.

(55) FIG. **3** shows two embodiments of the lock assembly **28** of the present invention. In FIG. **3**, the embodiments are shown side by side attached to a rung of the moving section. In reality, only one embodiment would be mounted in the center of the rung.

(56) FIGS. **4-10** show the operation of the first embodiment. A socket **34** is mounted within each of

the fixed (base) section rungs. A flip lock **36** is attached to one rung of the moving section. FIG. **4** shows the flip lock **36** in the position to which it returns when free. FIG. **5** shows the flip lock **36** is deflected as the base rung and fly rung begin to pass each other. FIG. **6** shows the flip lock **36** as it enters a socket **34**. FIG. **7** shows the flip lock **36** wedged in the socket **34**, which prevents the fly rung from further descending. FIG. **8** shows raising the fly rung withdraws the flip lock **36** from engagement with the socket **34**. FIG. **9** shows the flip lock **36** is returned to its spring-biased position after it has cleared the base rung. FIG. **10** shows the flip lock **36** deflects from the base rung to allow the fly rung to freely descend.

(57) A flip lock **36** is attached to one rung of the moving section. The flip lock **36** can be pivoted up or down but is biased by spring force to return to the position seen in FIG. **4** when free. Raising the moving section upward allows the stem **38** of the flip lock **36** to enter the socket **34**. At that point, lowering the moving section causes the stem **38** of the flip lock **36** to wedge into the socket **34** (FIGS. **5-7**). In this condition, the moving section is prevented from moving downward relative to the fixed section because the size and the shape of the socket **34** prevents the stem **38** from moving in the socket **34**. To unlock the moving section, it is raised until the flip lock **36** is above the fixed section rung (FIGS. **8** and **9**). At this point, the moving section can be lowered (FIG. **10**) as far as desired. It should be noted that when the stem **38** is moving down from above the fixed section rung, the stem **38** is cammed upwards by the fixed section rung as the step moves downward past the fixed section rung, so the stem **38** slides by the socket **34** and does not enter the socket **34**. It is only when the stem **38** is moving upwards, and it clears the lower portion of the fixed section rung and is forced under the action of the flip spring **44** back to its stable position can it enter into the socket **34** so that when the stem **38** is now in the socket **34** and the moving section is then lowered, does the stem **38** wedge into the socket **34**. The upper end of the stem **38**, at the bottom is angled inwards to facilitate the movement of the stem **38** out of the socket **34** when the moving section is lifted. The shape of the socket **34** is essentially flat along the top of the socket **34** so the stem **38** has essentially no place to move to come out of the socket **34** when a downward force is applied to the stem **38**, thus locking the fixed section and the moving section, while the rear of the socket **34** is more rounded to allow the upper end of the stem **38** clearance to come out of the socket **34** when the moving section is lifted.

(58) FIGS. **11-17** show the operation of the second embodiment of the lock assembly **128**. FIG. **11** shows the lock **46** and the retractor **48** in the position to which they are spring biased when free of any base rung **18**. FIG. **12** shows the lock **46** is deflected by the base rung **18** as the fly rung **26** is lifted. FIG. **13** shows the lock **46** has cleared the base rung **18** and the retractor **48** has been partially deflected by the base rung **18**. FIG. **14** shows the lock **46** has engaged the base rung **18** to prevent further motion downward as the fly rung **26** has been moved down and the retractor **48** is clear of the base rung **26**. FIG. **15** shows the retractor **48** is deflected by the base rung **18** as the fly rung **26** is lifted and the lock **46** is clear of the base rung **18**. FIG. **16** shows both the lock **46** and the retractor **48** are free and clear of the base rung **18** as the fly rung **26** has been lifted past the base rung **18**. FIG. **17** shows the retractor **48** has fitted into the slot **62** of the lock **46** because the base rung **18** has deflected the retractor **48** as the fly rung **26** is moved down, thus allowing the fly rung **26** to descend freely.

(59) A lock **46** and a retractor **48** of the lock assembly **128** are attached to one rung **26** of the moving section **20**. Both lock **46** and retractor **48** can be pivoted up and down but are biased by spring forces to return to the positions seen in FIG. **11**. FIGS. **12-14** show the locking sequence which is very similar to the locking assembly **28** of the first embodiment. A difference is that when the moving section **20** is locked (FIG. **14**), the lock **46** is braced against the top rear of the fixed section rung **18** by the first arm **58** of the lock **46** extending over the fixed section rung **18** and contacting the top flange **76** of the fixed section rung **18** and the second arm **60** of the lock **46** contacting and pressing against the web **70** of the fixed section rung **18**, so this embodiment does not require a socket **34**. In this condition, the moving section **20** is prevented from moving

downward relative to the fixed section **12** because the lock **46** has nowhere to move with respect to a downward force. To unlock the moving section **20**, it is raised until both the lock **46** and retractor **48** are above the fixed section rung **18** (FIG. **16**). Then as the moving section **20** is lowered, the first retractor arm **78** of the retractor **48** is cammed upward by the fixed section rung **18** it is moving past, causing the second retractor arm **80** to fit into the slot **62** of the lock **46**, which is now free and clear from any fixed section rung **18**, and retracts the lock **46** so that the moving section **20** can be lowered (FIG. **17**) as far as desired. Once the lowering of the moving section **20** has stopped, under the action of the retractor spring **66** and the lock spring **64** of the retractor **48** and the lock **46**, respectively, the retractor **48** and the lock **46** will return to their stable positions as long as there is no fixed section rung **18** blocking them.

(60) FIG. **18** shows the lock assembly **28** for the first embodiment. There is a frame **68** that is riveted or bolted to the desired rung **26** of the moving section **20**. The flip axle **42** is mounted to the frame **68** and the flip lock **36** is mounted to the flip axle **42**. There is a flip spring **44** positioned about the flip axle **42** with a top end and a bottom end of the flip spring **44** extending outwards. Between the top end and the bottom end of the flip spring **44** is a flip lock stop **74** extending from the flip lock **36** and a frame stop **72** extending from the frame **68**. The flip lock stop **74** and the frame stop **72** bias the respective spring as they move, as shown in FIGS. **19-21**.

(61) FIG. **19** a cross-sectional side view showing the flip lock stop **74** and the frame stop **72** and the flip lock **36** when the flip lock **36** is free and clear of any base rung **18**. FIG. **20** is a cross-sectional side view of the flip lock **36** when it has been deflected upwards by a base rung **18**. The flip lock stop **74** has pushed against the top of the flip spring **44** and the frame stop **72** has maintained the bottom of the flip spring **44** in position, thus causing a biasing force by the top of the spring pushing downwards against the flip lock stop **74**. FIG. **21** is a cross-sectional side view of the flip lock **36** when it has been deflected downwards by a base rung **18**. The flip lock stop **74** has pushed against the bottom of the flip spring **44** and the frame stop **72** has maintained the top of the flip spring **44** in position, thus causing a biasing force by the bottom of the spring upwards against the flip lock **36**.

(62) FIG. **22** shows the lock assembly **128** of the second embodiment. There is a frame **68** that is riveted or bolted to the desired rung **26** of the moving section **20**. There is a lock axle **50** having the lock **46** mounted to the frame **68** and a retractor axle **54** having the retractor **48** mounted to the frame **68**. There is a lock spring **64** positioned about the lock axle **50** and a retractor spring **66** positioned about the retractor axle **54**. Between the top end and the bottom end of the lock spring **64** is a lock stop **73** and a frame stop **72**, and between the top end and the bottom end of the retractor spring **66** is a retractor stop **75** and another frame stop **72**.

(63) The frame stops **72** and the lock stop **73** and the retractor stop **75** bias the respective springs as they move, as shown in FIGS. **22-25**. FIG. **23** is a cross-sectional side view of the lock **46** and the retractor **48** and the frame stops **72** and the lock stop **73** and the retractor stop **75** when the lock **46** and the retractor **48** are free and clear of any base rung **18**.

(64) FIG. **24** is a cross-sectional side view of the lock **46** and the retractor **48** when they had been deflected upwards by a base rung **18**. The lock stop **73** has pushed against the top of the lock spring **64** and the frame stop **72** has maintained the bottom of the lock spring **64** in place, causing a biasing force of the top of the lock spring **64** against the lock stop **73**. Similarly, the retractor stop **75** has pushed against the top of the retractor spring **66** and the frame stop **72** has maintained the bottom of the retractor spring **66** in place, causing a biasing force of the top of the retractor spring **66** against the retractor stop **75**.

(65) FIG. **25** is a cross-sectional side view of the lock **46** and that retractor **48** when they have been deflected downwards by a base rung **18**. The lock stop **73** has pushed against the bottom of the lock spring **64** and the frame stop **72** has maintained the top of the lock spring **64** in place, causing a biasing force of the bottom of the lock spring **64** against the lock stop **73**. Similarly, the retractor stop **75** has pushed against the bottom of the retractor spring **66** and the frame stop **72** has

maintained the top of the retractor spring **66** in place, causing a biasing force at the bottom of the retractor spring **66** against the retractor stop **75**.

(66) Referring now to the drawings wherein like reference numerals refer to similar or identical parts throughout the several views, and more specifically to FIGS. **26-33** thereof, there is shown a climbing product **10** whose sections are extendable. The climbing product **10** comprises a base section **12** having a first base rail **14** and a second base rail **16** in parallel and spaced relationship with the first base rail **14** and base rungs **18** attached to and between the first and second base rails **14, 16**. The climbing product **10** comprises a fly section **20** having a first fly rail **22** and a second fly rail **24** in parallel and spaced relationship with the first fly rail **22** and fly rungs **26** attached to and between the first and second fly rails **22, 24**. The fly section **20** in sliding engagement with the base section **12**. The climbing product **10** comprises a lock assembly **28** having a lock fly portion **30** directly attached to the fly section **20**. The lock fly portion **30** directly engages with the base section **12** to lock the base section **12** with the fly section **20**. The lock fly portion **30** has a plunger **39**. The lock fly portion **30** disengages from the base portion by moving the plunger **39**.

(67) The lock assembly **28** may include a lock base portion **32** directly attached to the base section **12**. The lock fly portion **30** directly engages with the lock base portion **32** to lock the fly section **20** to the base section **12**. The lock fly portion **30** disengages from the lock base portion **32** by moving the plunger **39**. The lock base portion **32** may include a socket **34** disposed in one of the base rungs **18**, and the lock fly portion **30** includes a flip lock **36** which fits into the socket **34** to lock the fly section **20** to the base section **12** so the fly section **20** cannot move downwards relative to the base section **12** when the fly section **20** is locked to the base section **12**. The flip lock **36** disposed in one of the fly rungs **26**.

(68) The flip lock **36** may have a stem **38** which fits into the socket **34** to lock the fly section **20** to the base section **12**, an anchor **40** from which the stem **38** extends, a flip axle **42** extending through the anchor **40** about which the anchor **40** with the stem **38** rotates, and a flip spring **44** which biases the stem **38** into a stable position where the stem **38** extends essentially perpendicularly toward the first base rail **14** from the first fly rail **22**. The flip lock **36** may have a locking recess **120** into which the plunger **39** moves. When the plunger **39** engages the recess, the flip lock **36** is prevented from rotating about its axis. The flip lock **36** may include a blocker **122**. The plunger **39** being able to engage the flip lock **36** to prevent the flip lock's rotation when the blocker **122** is in a first state and does not block the plunger **39** from engaging with the locking recess **120**, and the plunger **39** being prevented from engaging the flip lock **36** by the blocking action of the blocker **122** in a second state and the plunger **39** is prevented from engaging with the locking recess **120**.

(69) The lock assembly **28** may include a lock lever **112** engaged with the plunger **39** and a lock lever handle **116** directly connected to the lock lever **112**. The lock lever handle **116** is moved from its down position to its up position by directly pulling on the lock lever handle **116** which moves the plunger **39** out from the locking recess **120**. The lock assembly **28** may include a pulley **124** mounted to the lock lever **112**, and including a hoist rope **126** which goes around the pulley **124**. By increasing tension in the hoist rope **126** the lock lever **112** is moved from its down position to its up position which moves the plunger **39** out from the locking recess **120**.

(70) The present invention pertains to a method for using a ladder whose sections are extendable. The method comprises the steps of moving the ladder to a desired position. The ladder comprises a base section **12** having a first base rail **14** and a second base rail **16** in parallel and spaced relationship with the first base rail **14** and base rungs **18** attached to and between the first and second base rails **14, 16**. The ladder comprises a fly section **20** having a first fly rail **22** and a second fly rail **24** in parallel and spaced relationship with the first fly rail **22** and fly rungs **26** attached to and between the first and second fly rails **22, 24**. The fly section **20** in sliding engagement with the base section **12** the ladder comprises a lock assembly **28** having a lock fly portion **30** directly attached to the fly section **20**. The lock fly portion **30** directly engages with the base section **12** to lock the base section **12** with the fly section **20**. The lock fly portion **30** has a

plunger **39**. The lock fly portion **30** disengages from the base portion by moving the plunger **39**. There is the step of moving the plunger **39** so the lock fly portion **30** disengages from base portion. There is the step of sliding the fly section **20** along the base section **12** to extend the ladder to a desired length. There is the step of locking the fly section **20** to the base section **12** with the lock portion at the desired length.

(71) The present invention pertains to a method for constructing a ladder whose sections are extendable. The method comprises the steps of attaching a lock fly portion **30** of the lock assembly **28** directly to a fly section **20** of the ladder. The fly section **20** having a first fly rail **22** and a second fly rail **24** in parallel and spaced relationship with the first fly rail **22** and fly rungs **26** attached to and between the first and second fly rails **22**, **24**. The fly section **20** in sliding engagement with the base section **12**. The lock fly portion **30** has a plunger **39**. There is the step of attaching a lock base portion **32** of the lock assembly **28** directly attached to the base section **12**. The lock fly portion **30** directly engages with the lock base portion **32** to lock the fly section **20** to the base section **12**. The lock fly portion **30** disengages from the lock base portion **32** by moving the plunger **39**. The base section **12** having a first base rail **14** and a second base rail **16** in parallel and spaced relationship with the first base rail **14** and base rungs **18** attached to and between the first and second base rails **14**, **16**.

(72) This invention is an additional feature, and an alternative embodiment for the locking assembly with the flip lock **36** described above. This invention prevents inadvertent unlocking of the Extendable Section Lock by requiring a deliberate action by the user. It is an additional feature that, for instance, may be introduced to the apparatus shown in FIGS. **26-35**, and **43-46**, and described in this application, specifically embodiment **1**, thereof.

(73) This invention prevents the unintended extension of a ladder section, which equates to increased user safety and convenience.

(74) FIGS. **26** and **27** show a ladder known as an NGMF (next generation multi form) ladder. This ladder can be folded and opened like a step ladder and also be put into a straight ladder configuration like an MT ladder. Also like an MT ladder, the NGMF has outer sections attached to its front and rear inner sections. These outer sections may be extended relative to the inner sections to increase the height of the ladder. The NGMF shown uses conventional J locks on its front section. The rear section uses the flip lock **36** invention which is mounted within the lowest rung of the rear inner section.

(75) FIG. **28** is a detailed view showing the inner rung with the flip lock **36** engaged with one of the outer rungs. Also shown is a hoisting rope **126** whose function will be described later. There is a hole **161** in the top surface of the fly rung **26** through which the rope **126** extends and wraps around the pulley **124**.

(76) FIG. **29** shows just the inner rung with flip lock **36** and an outer rung.

(77) FIG. **30** shows another view of just the inner rung with flip lock **36**.

(78) FIG. **31** shows another view of a typical outer rung with a recess for flip lock **36** to engage. These recesses are identical regardless of the length of the particular outer rung.

(79) FIG. **32** shows the components of this invention as seen if the inner rung were transparent. Important to note is that the lock lever pivot **114** and the crank pivot **110** are fixed to the inner rung. The lock lever pivot **114** and the crank pivot **110** can rotate about their respective pivots.

(80) FIG. **33** shows a closeup of FIG. **32** with some additional components made invisible. The key parts of this invention are the flip lock **36**, blocker **122**, and plunger **39**. The essence of this invention is the plunger **39** being able to engage the flip lock **36** to prevent its rotation in certain conditions, and, the plunger **39** being prevented from engaging the flip lock **36** by the blocking action of the blocker **122** in other conditions.

(81) FIG. **34** shows the relationship between the flip lock **36** and the blocker **122**. The blocker **122** pivot is fixed to the flip lock **36**. The blocker **122** can rotate CCW and CW relative to the flip lock **36**. The blocker **122** is spring biased towards its extreme CW position as seen in FIGS. **37**, **38** and

39.

(82) FIG. 35 shows a portion of the flip lock 36. The flip lock 36 has a locking recess 120 into which the plunger 39 can go. When the plunger 39 engages this recess, the flip lock 36 is prevented from rotating about its axis.

(83) FIG. 36 shows the conditions under which the plunger 39 is able to engage the plunger 39 locking recess 120 in the flip lock 36: the flip lock 36 is horizontal and the blocker 122 is rotated to its extreme CCW position relative to the flip lock 36.

(84) FIGS. 37, 38 and 39 show how the plunger 39 (axis shown by +symbol) is prevented from engaging the plunger 39 locking recess 120 (shown as a dashed line) whenever the blocker 122 is in its extreme CW position relative to the flip lock 36.

(85) FIGS. 40 and 41 show the function of the lock lever 112 and the plunger 39. When the right end of the lock lever 112 is down as in FIG. 40, the plunger 39 is in its left or engaged position with the locking recess 120. (This assumes that the blocker 122 is not preventing the plunger 39 from entering the plunger 39 locking recess 120 in the flip lock 36.)

(86) When the right end of the lever is lifted as in FIG. 41, the plunger 39 is moved to its right or disengaged position from the locking recess 120. The plunger 39 is moved by the action of the stud 118 on the lock lever 112 which causes the crank 108 to rotate.

(87) The lock lever 112 is moved from its down position to its up position by the user either directly pulling on the lock lever handle 116 which is directly connected to the lock lever 112, or by increasing the tension in the hoist rope 126 which goes around the pulley 124 mounted on the right end of the lock lever 112.

(88) FIG. 42 shows how the contact between the blocker 122 against the outer rung positions the blocker 122 CCW relative to the flip lock 36 as seen in FIG. 36. Thus, FIG. 42 shows the only situation in which the blocker 122 allows the plunger 39 to prevent the rotation of the flip lock 36: the flip lock 36 is fully engaged with the recess in outer rung and the lock lever 112 is allowed to move to its down position.

(89) FIG. 43 shows the situation whenever the flip lock 36 is NOT engaged in the recess in an outer rung: the blocker 122 prevents the plunger 39 engaging the flip lock 36 so the flip lock 36 is free to move in the manner of the previously disclosed flip lock 36 invention.

(90) As shown in FIGS. 32, 33, 40 and 41, when the hoist rope 126 wrapped around the pulley 124 is pulled taught to lift the fly section 20 relative to the base section 12, or the lock lever handle 116 is lifted, the lock lever 112 rotates upwards about the lock lever pivot 114. As the lock lever 112 moves upwards, a stud 118 pushes against a crank 108, as shown in FIG. 40, causing the crank 108 to pivot around the crank pivot 110 and pull the plunger 39, which is directly connected to the crank 108, out from the locking recess 120 and be completely withdrawn from the flip lock 36, as shown in FIG. 41. With the plunger 39 completely withdrawn from the flip lock 36, the flip lock 36 is free to rotate about the flip lock axis 106. When the rope 126 is released, or the lock lever handle 116 is let go, the lock lever pivot 114 moves back down about the lock lever pivot 114, causing the stud 118 to move back away from the crank 108 and the crank 108 to rotate clockwise about the crank pivot 110, which causes the plunger 39 to move forward to try to enter the locking recess 120 again. If the blocker 122 is in the way of the locking recess 120, the blocker 122 prevents the plunger 39 from entering the locking recess 120 and re-engaging with the flip lock 36 to lock the flip lock 36 in place.

(91) When the fly section 20 is locked with the base section 12, the flip lock 36 has its stem 38 in the socket 34 of the desired base rung with respect to the height of the fly section 20 relative to the base section 12. In this event, the plunger 39 is disposed in the locking recess 120 preventing the flip lock 36 from being able to rotate, as shown in FIG. 36 and FIG. 42, and thus preventing the flip lock 36 from coming out of the locking recess 120 and consequently preventing the fly section 20 moving relative to the base section 12.

(92) When it is desired to reconfigure the position of the fly section 20 relative to the base section

12, either the hoist rope **126** is pulled, or the handle is lifted to cause the plunger **39** to withdraw from the locking recess **120** and free the flip lock **36** to rotate, as explained above. The fly section **20** is then lifted relative to the base section **12**, which causes the stem **38** of the flip lock **36** to withdraw from the socket **34**. This is able to occur because the bottom front surface of the stem **38** is curved, and as the fly section **20** is lifted, it causes the stem **38** of the flip lock **36** to go with the fly section **20** and slide out unobstructed from the socket **34**. Because the flip lock **36** is now free of the socket **34**, the fly section **20** is able to continue to be lifted unimpeded. Once the blocker **122** attached to the flip lock **36** clears the base rung the flip lock **36** was just in, the nose **137** of the blocker **122** is no longer displaced by the base rung, and the blocker **122** is biased back to its stable position by the flip lock spring, causing the blocker **122** to rotate clockwise about the blocker pivot **102** over the locking recess **120** so the plunger **39** cannot slide into the locking recess **120**, as shown in FIG. **43**, because the plunger axis **104** of the plunger **39** that aligns with the locking recess is blocked by the blocker **122**. It should be noted that only when the nose **137** of the blocker **122** is deflected by the base rung and rotates counter clockwise about the pivot blocker **102**, can the plunger **39** engage and move into the locking recess **120**. In all other positions, the flip lock spring biases the blocker **122** into a position that blocks the plunger **39** from sliding into the locking recess **120**.

(93) Even when the rope **126** is tensioned or the handle is pulled up and the plunger **39** has withdrawn from the locking recess **120**, the fly section **20** is prevented by the flip lock **36** inserted in the socket **34** to be moved directly down. This is because the top surface of the stem **38** is flat, so when the fly section **20** is moved down, the top surface of the stem **38** hits the top of the socket **34** and has nowhere to go, as explained above in regard to the first embodiment of the flip lock **36**. In this instance, the flip lock **36** cannot slide out of the socket **34**. Accordingly, only when the fly section **20** is lifted upwards, can the flip lock **36** withdraw from the socket **34**, and the plunger **39** has been withdrawn.

(94) Once the fly section **20** has been lifted and the flip lock **36** has cleared the base rung, the fly section **20** can be moved downward. As the fly section **20** moves down, the flip lock **36** and the nose **137** strike the top of the base rung and are deflected upwards, as shown in FIG. **38**. It should be noted that the blocker **122** pivots forward CW about the blocker **122** pivot and the flip lock axle moves to the rightmost position of the blocker slot **139** because the nose **137** is no longer being deflected by the base rung. As the fly section **20** continues to move down, the stem **38** of the flip lock **36** slides past the socket **34** of the base rung and continues moving down until it clears the base rung, where the flip lock **36** and the blocker **122** will revert to the position shown in FIG. **37**.

(95) When the fly section **20** is being lifted upwards, the stem **38** of the flip lock **36** and the nose **137** of the blocker **122** will strike the bottom of the base rung in the deflected downwards by the base rung, as shown in FIG. **39**. As the fly section **20** continues to be moved upwards, the stem **38** and the nose **137** will move past the socket **34** of the base rung and continue upwards until the flip lock **36** and the blocker **122** cleared the base rung, whereupon they will revert back to the position shown in FIG. **37**. Only one fly rung **26** has the flip lock **36** and blocker **122** arrangements, but at least some if not all of the base rungs **18** will have a socket **34** so the fly section **20** can be positioned at a desired height relative to the base section **12** by having the flip lock **36** for the into the desired socket **34**.

(96) For the flip lock **36** to be repositioned in a socket **34** of another base rung, this can only occur as the fly section **20** is moving up relative to the base section **12**. Instead of the flip lock **36** passing past a socket **34**, as the fly section **20** is moved upward, as explained above, the fly section **20** is stopped from continuing to move upward about where the fly rung **26** with the flip lock **36** is essentially adjacent to the base rung. At this point, the top surface of the flip lock **36** will be inserted slightly into the socket **34**, with the flip lock **36** and buttress positioned essentially as shown in FIG. **39**. When the top of the flip lock **36** hits the top surface of the socket **34**, there may be heard a clicking or contact sound to alert the user that the flip lock **36** is properly position. Then,

the fly section **20** is moved down slightly so the stem **38** of the flip lock **36** slides fully into the socket **34** and the nose **137** of the blocker **122** is deflected downwards by the base rung and takes the position shown in FIG. **36**. As explained above, the configuration of FIG. **36** has the blocker **122** opening now aligned with the locking recess **120**, so the plunger **39** now slides into the locking recess **120** and engages with the flip lock **36**, to lock the flip lock **36** in place with the socket **34** of the base rung.

(97) FIG. **44** shows the routing of the hoist rope **126** between the base section **12** and the fly section **20**. A first end **152** of the rope **126** is attached to the top of the outer/base section **12** of the rear section **142**. From there, the rope **126** extends down the base rail of the rear section **142**, where the rope **126** passes around the pulley **124** on the lock lever **112**. From there, the rope **126** extends along the base rail and the fly rail above the base rail to the hinge **148** where it is routed to the front section **140**. At the front section **140** extends down along the fly rail of the front section **140** to the base rail below the fly rail of the front section **140** where the second end **154** of the rope **126** is attached to the lowest fly rung **26** on the inner/fly section **20** of the front section **140**.

(98) FIG. **45** shows the ladder in the fully extended straight extension ladder position from the non-climbing side.

(99) FIG. **46** shows the ladder in the fully extended straight extension ladder position from the climbing side.

(100) FIGS. **47A-47E** show multiple views of the lock lever **112** and lock lever handle **116** in place on the fly rung **26**. The lock lever **112** and lock lever handle **116** are mounted external to the back wall **121** of the fly rung **26**. The pulley **124** on the lock lever **112** however intrudes into the interior of the fly rung **26**. FIG. **47A** is a perspective view of the top of the fly rung **26** and lock assembly **28**. FIG. **47B** is a right-side view of the fly rung **26** and lock assembly **28**. FIG. **47C** is a backside/right side view of the fly rung **26** and lock assembly **28**. FIG. **47D** is a top view of the fly rung **26** and lock assembly **28**. FIG. **47E** is a front side, upside down of the fly rung **26** and lock assembly **28**. FIG. **47F** is a left side view of the fly rung **26** and lock assembly **28**. As can be seen from these figures, the lock assembly **28** essentially fits within and is protected by the fly rung **26**. The flip lock **36** and the handle **116** of the lock assembly **28** are basically the only components which extend beyond the fly rung **26**.

(101) FIG. **48** is a perspective view of the ladder with the fly sections extended. FIG. **49** is a perspective view of the flip lock **36** with the blocker **122** and the flip lock spring **145**, which biases the blocker **122**, in place in the flip lock **36** and adjacent the blocker **122**.

(102) The ladder **10** may be a standard extension ladder having a fly section **20** and a base section **12**. Alternatively, the ladder **10** may be a multipurpose ladder having a front section **140** and a rear section **142**, as shown in FIG. **26** and FIG. **27**, in an unextended step ladder configuration, or FIG. **48** in an extended step ladder configuration, or in FIGS. **45** and **46** in an unextended extension ladder configuration. As a multipurpose ladder the front section may have J locks **144** for locking the fly section **20** and the base section **12** together at a desired length, and the rear section **142** having the lock assembly **28** described herein for locking the fly section **20** and the base section **12** of the rear section **142** at a desired length. If desired, the front section **140**, instead of having J locks **144**, may use the lock assembly **28** for locking the fly section **20** and base section **12** together.

(103) As a multipurpose ladder **10**, the front section **140** also has a base section **12** having a first base rail **14** and a second base rail **16** with base rungs **18** attached to them and disposed between them, as described above regarding the base section **12** and the fly section **20**. The front section **140** also has a fly section **20** having a first fly rail **22** and a second fly rail **24** with fly rungs **26** attached to them and disposed between them, as described above regarding the base section **12** and the fly section **20**. The base rungs **18** attached to the outside flanges of the first and second base rails, and the fly rungs **26** are attached to the inside of the web of the first and second fly rails so as not to interfere with the movement of the fly section **20** relative to the base section **12**.

(104) The first and second fly rails of the front section are attached with rivets or fasteners to a

ladder top **146**. Hinges **148** are attached by rivets or fasteners to the first and second fly rails of the front section adjacent the ladder top **146**. The tops of the fly rails of the rear section **142** are attached to the hinges **148**. The hinges **148** allow the rear section **142** to rotate relative to the front section **140**. As shown in FIGS. **20** and **21**, the rear section **142** has rotated about 180° relative to the front section **140** form the extension ladder position. In the extension ladder position, the first fly rail **22** of the rear section **142** and the second fly rail **24** of the rear section **142** fit into channels **150** in the ladder top **146** to allow the rear section **142** to align with the front section **140**, although slightly offset due to the connection of the rear section fly rails being attached to the hinges **148**. (105) Although the invention has been described in detail in the foregoing embodiments for the purpose of illustration, it is to be understood that such detail is solely for that purpose and that variations can be made therein by those skilled in the art without departing from the spirit and scope of the invention except as it may be described by the following claims.

Claims

1. A ladder comprising: a base section including a first base rail and a second base rail in a spaced relationship with the first base rail and a plurality of base rungs attached to and extending between the first base rail and the second base rail; a fly section including a first fly rail and a second fly rail in a spaced relationship with the first fly rail and a plurality of fly rungs attached to and extending between the first fly rail and the second fly rail, the fly section in sliding engagement with the base section; and a lock assembly including a lock base portion attached to at least one of the plurality of base rungs of the base section and a lock fly portion attached to the fly section, the lock fly portion engages with the lock base portion to lock the base section with the fly section, wherein the lock base portion and the lock fly portion directly engage with one another when the fly section is being extended relative to the base section and the lock base portion permitting relative movement of the fly section when the fly section is being retracted.
2. The ladder of claim 1 wherein the lock base portion includes a socket disposed in one of the plurality of base rungs, and the lock fly portion includes a flip lock which fits into the socket to lock the fly section to the base section to prevent inadvertent unlocking and retraction of the fly section or the base section relative to one another when the fly section is locked, the flip lock disposed in one of the plurality of fly rungs.
3. The ladder of claim 2 wherein the flip lock includes a stem which fits into the socket to lock the fly section to the base section, an anchor from which the stem extends, a flip axle extending through the anchor about which the anchor including the stem rotates, and a flip spring which biases the stem into a stable position where the stem extends substantially perpendicular toward the first base rail from the first fly rail.
4. The ladder of claim 2 wherein the flip lock includes a beveled edge to permit movement into the socket in one direction.
5. The ladder of claim 2 wherein the fly section is permitted to move when the flip lock is biased by and outside of the socket.
6. The ladder of claim 2 wherein the lock fly portion disengages from the lock base portion by moving the one of the plurality of fly rungs past the socket and retracting the fly section.
7. The ladder of claim 2 wherein the flip lock travels past the socket before settling in the socket to lock the lock fly portion to the lock base portion.
8. The ladder of claim 1 wherein the lock fly portion includes a lock which engages with one of the plurality of base rungs.
9. The ladder of claim 8 wherein the lock fly portion includes a retractor disposed adjacent the lock which engages the lock and maintains the lock in a position away from the plurality of base rungs as the fly section is moved relative to the base section.
10. The ladder of claim 9 wherein the lock fly portion includes a lock axle extending through the

lock about which the lock rotates and a retractor axle extending through the retractor about which the retractor rotates.

11. The ladder of claim 10 wherein the lock includes an anchor, a first arm extending from the anchor and a second arm in a spaced relation with the first arm, where the first arm is longer than the second arm and where the second arm extends from the anchor, the first arm and the second arm and the anchor together form a substantially C shape and define a slot between the first arm in the second arm, the lock fly portion includes a lock spring which biases the lock into a stable position where the first arm and the second arm extend toward the first base rail, the lock fly portion includes a retractor spring which biases the retractor into a stable position where the retractor extends toward the first base rail.

12. The ladder of claim 1 wherein the fly portion includes a stem that engages with a socket of the base portion and wherein the stem comprises a flat top surface and an angled bottom surface.

13. The ladder of claim 1 wherein each of the plurality of the base rungs includes the lock base portion having a socket therein.

14. The ladder of claim 1 wherein movement of the lock fly portion to a position past the lock base portion unlocks the lock fly portion from the lock base portion and permits a disengagement of the fly section from the base section.

15. A ladder comprising: a base section including a first base rail and a second base rail, the second base rail in a spaced relation to the first base rail; a fly section including a first fly rail and a second fly rail, the second fly rail in a spaced relation to the first fly rail; a plurality of base rungs attached to and extending between the first base rail and the second base rail; a plurality of fly rungs attached to and extending between the first fly rail and the second fly rail; and a locking mechanism including a fly lock portion coupled to at least one of the plurality of fly rungs and a base lock portion coupled to at least one of the plurality of base rungs, wherein the fly lock portion and the base lock portion selectively couple to one another to lock the base section and the fly section together to prevent inadvertent relative movement and retraction of the base section or the fly section relative to one another.

16. The ladder of claim 15, wherein the base lock portion includes a socket disposed within the at least one of the plurality of base rungs, wherein the fly lock portion includes a movable stem coupled to the at least one of the plurality of fly rungs, and wherein the fly rail and base rail are locked to one another via the movable stem engaging the socket.

17. The ladder of claim 15, wherein the base lock portion includes a socket disposed within the at least one of the plurality of base rungs.

18. The ladder of claim 17, wherein the socket includes a substantially flat top wall and a substantially rounded lower wall portion.

19. The ladder of claim 15, wherein the fly lock portion includes a flip lock coupled to the at least one of the plurality of fly rungs.

20. The ladder of claim 19, wherein the flip lock is coupled to the at least one of the plurality of fly rungs via an anchor and axle.

21. The ladder of claim 19, wherein the flip lock is coupled to the at least one of the plurality of fly rungs via a frame.

22. The ladder of claim 19, wherein the flip lock includes a substantially flat top surface, a substantially flat bottom surface, and an end surface, wherein at least a portion of the end surface is angled.

23. The ladder of claim 19, wherein the flip lock includes a flip lock spring to bias the flip lock.

24. The ladder of claim 15 wherein the fly portion includes a stem that engages with a socket of the base portion and wherein the stem comprises a flat top surface and an angled bottom surface.

25. The ladder of claim 15 wherein each of the plurality of the base rungs includes the lock base portion having a socket therein.

26. A ladder comprising: a base section including a first base rail and a second base rail, the second

base rail in a space relation to the first base rail; a fly section including a first fly rail and a second fly rail, the second fly rail in a spaced relation to the first fly rail; a plurality of base rungs attached to and extending between the first base rail and the second base rail; a plurality of fly rungs attached to and extending between the first fly rail and the second fly rail; and a locking mechanism including a fly lock portion coupled to at least one of the plurality of fly rungs and a base lock portion coupled to at least one of the plurality of base rungs, wherein the base lock portion includes a socket disposed within the at least one of the plurality of base rungs, wherein the fly lock portion includes a flip lock coupled to the at least one of the plurality of fly rungs, the flip lock including a lock and a retractor, the retractor sized to engage at least a portion of the lock, and wherein the retractor includes a retractor spring to bias the retractor, wherein movement of the fly section relative to the base section causes the retractor to engage the lock and in turn engage the flip lock to the socket preventing inadvertent unlocking and retraction of the base section or the fly section relative to one another.

27. The ladder of claim 26, wherein the lock further comprises a lock spring to bias the lock.

28. The ladder of claim 26, wherein the lock and the retractor are connected via a frame.
